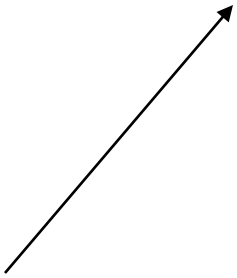
Compiled Problems – Senior Division

1. 11-12 C1 Assembly Language

What is the final value of N when the program is run?

```
A     DC      32          LOAD    C
B     DC      2           STORE   A
N     DC      0           BG      TOP
TOP   LOAD    A           LOAD    N
      DIV     B           SUB     =1
      STORE   C           STORE   N
      LOAD    N           PRINT   N
      ADD     =1          END
      STORE   N
```



2. 12-13 C1 Assembly Language

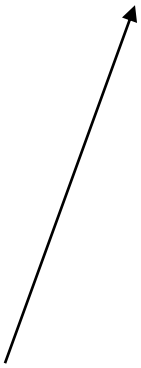
What is the final value of S that is printed when the program is run?

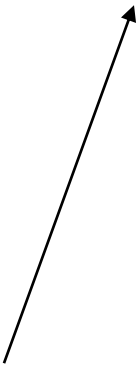
```
S     DC      0
N     DC      10
TOP   LOAD    N
      MULT    N
      ADD     S
      STORE   S
      LOAD    N
      SUB     =1
      STORE   N
      BG      TOP
      PRINT   S
      END
```

3. 13-14 C1 Assembly Language

What is the final value of E when the program is run?

```
A     DC      10          T     LOAD    A           P     LOAD    A
B     DC      2                 SUB     D                 SUB     B
C     DC      5                 BE      U                 STORE   A
D     DC      8           V     LOAD    A                 BU      R
      LOAD    A                 DIV     B           S     LOAD    B
      SUB     B                 ADD     C                 ADD     C
      BG      P                 MULT    D                 STORE   B
R     LOAD    B                 STORE   E                 BU      T
      MULT    C                 LOAD    E           U     LOAD    D
      SUB     D                 PRINT   E                 SUB     D
      BL      S                 END                       STORE   D
                                                          BU      V
```





4. 14-15 C1 Assembly Language

What is printed when the program is run?

```
X       DC      21      LOAD    K               BU      DONE
Y       DC      20      ADD     L       RSLT    LOAD    T
Z       DC      23      STORE   M               PRINT   T
T       DC      1       LOAD    Z       DONE    END
F       DC      0       MULT    Z
        LOAD    X       STORE   N
        MULT    X       LOAD    M
        STORE   K       SUB     N
        LOAD    Y       BE      RSLT
        MULT    Y       LOAD    F
        STORE   L       PRINT   F
```

5. 15-16 C1 Assembly Language

What is printed when the program is run?

```
A       DC      2       LOAD    D
B       DC      1       ADD     = 1
C       DC      1       STORE   D
D       DC      1       LOAD    C
TOP     LOAD    B       SUB     = 100
        MULT    A       BL      TOP
        STORE   B       PRINT   D
        LOAD    C       END
        ADD     B
        STORE   C
```